

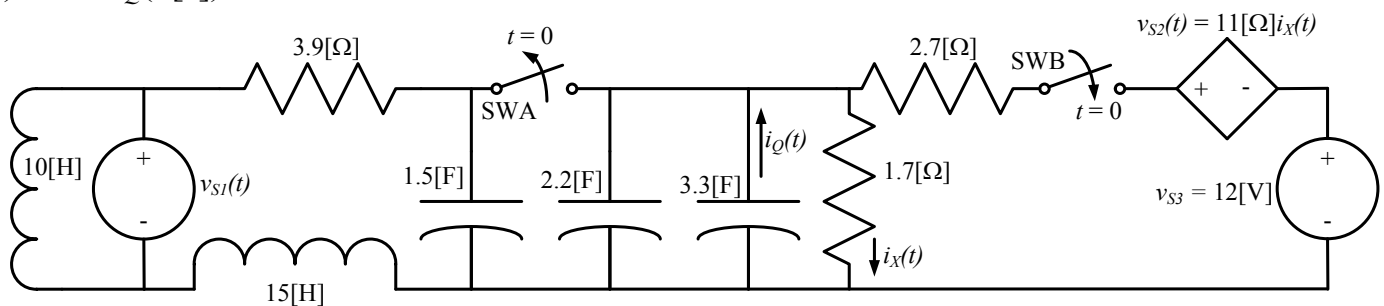
Name: Anaya Sco (please print)

ECE 2202 – HW 7

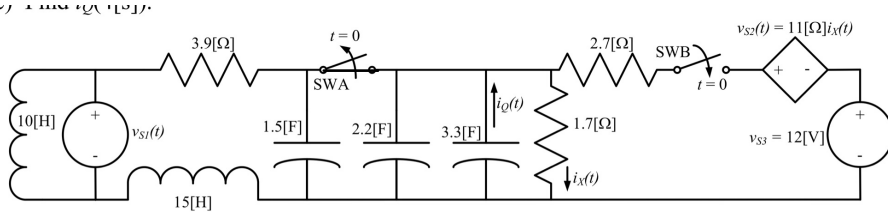
1.

The switch SWA had been closed, and switch SWB had been open for a long time before $t = 0$. Then at $t = 0$, switch SWA opened, and switch SWB closed.

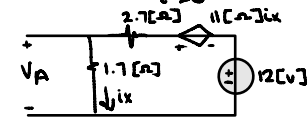
- Find the power delivered by $v_{SI}(t)$ at $t = 0^-$.
- Find $i_X(2[s])$.
- Find $i_Q(4[s])$.



(solution in [Final Exam Spring 2016, problem 5](#))



continue Part b)



KCL

$$12[V] = 1.7[\Omega]i_X + 2.7[\Omega]i_X - 11[\Omega]i_X$$

$$i_X = -1.88[A]$$

$$V_A = i_X \cdot 1.7[\Omega] = -3.09[V]$$

$$V_A(t) = (-3.09 + \frac{2.96}{1.7}e^{-t/3.826})[V]$$

from the inverse transform.

$$V_A(t) = 4.00[V] \cos(70[\text{mrad/s}]t - 42.3^\circ)$$

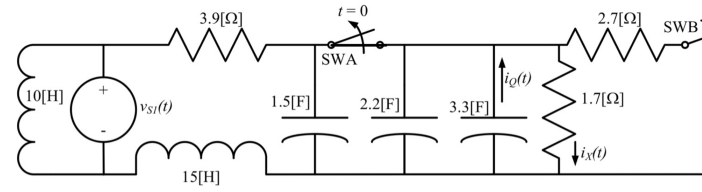
$$V_A(0) = 4.00[V] \cos(-42.3^\circ) = 2.96[V]$$

$$i_X(2cs) = \frac{V_A(2cs)}{1.7[\Omega]} = \frac{(-3.09 + 6.05 e^{2/3.826})[V]}{1.7[\Omega]}$$

$$i_X(2cs) = 4.19[A]$$

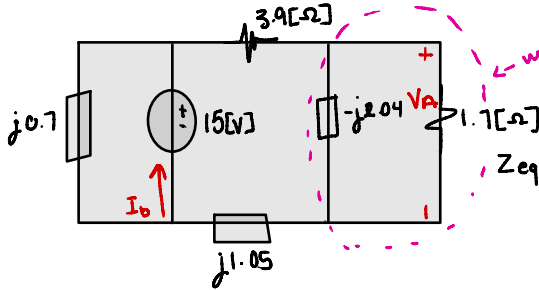
Set up

t < 0



$$C_{eq} = 1.5[F] + 2.2[F] + 3.3[F] = 7.0[F]$$

Convert: $V_{S1}(t) = 15[V] \cos(70[\text{mrad/s}]t)$
 $Z_L = j\omega L \rightarrow j(70e-3)(10[H]) \rightarrow j0.7[\Omega]; j(70e-3)(15[H]) = j1.05[\Omega]$
 $Z_C = \frac{j}{\omega C} \rightarrow \frac{j}{(70e-3)(7[F])} \rightarrow -j2.04[\Omega]$



we can find Zeq from this area for V_A

$$Z_{eq} = 1.7[\Omega] \parallel -j2.04[\Omega]$$

$$\frac{(1.7 \cdot -j2.04)[\Omega]}{1.7[\Omega] - j2.04[\Omega]} = 1.31[\Omega] \angle -39.8^\circ$$

$$V_A = 15[V] \left(\frac{1.31 \angle -39.8^\circ}{3.9 + j1.05 + 1.31 \angle -39.8^\circ} \right) = 4[V] \angle -42.3^\circ$$

I believe we don't include j0.7 cause we are seeing the circuit from V_A's pov.

$$I_0 = \frac{15[V]}{j0.7[\Omega]} + \frac{15[V]}{1.9 + j1.05 + 1.31 \angle -39.8^\circ} = 21.8 \angle -81.9^\circ[A]$$

in time domain: $21.8[A] \cos(70[\text{mrad/s}]t - 81.9^\circ)$

Part A) Find the power delivered by $V_{S1}(t)$ at $t=0$

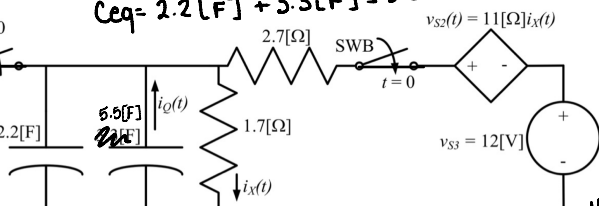
$$P_{del, V_{S1}} = V_{S1}(t) \cdot I_0(t)$$

$$(15[V] \angle 0^\circ)(21.8[A] \angle -81.9^\circ) = 46.1[W]$$

Part b) Find $i_X(2cs)$

t > 0

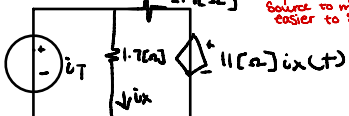
$$C_{eq} = 2.2[F] + 3.3[F] = 5.5[F]$$



$$i_X = \frac{1[V]}{1.7[\Omega]} = 0.588[A]$$

We need to find Req.

We take out the Ceq and independent source to make it easier to solve for Req.



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$$i_T = \frac{1[V]}{1.7[\Omega]} + \frac{1[V] - 11[\Omega]i_X}{2.7[\Omega]}$$

$$= -1.438[A]$$

$$R_{eq} = \frac{1[V]}{-1.438[A]} \rightarrow -0.695$$

$$Z = R_{eq} C_{eq} \rightarrow (5.5[F])(-0.695[\Omega]) = -3.825[cs]$$

Part c) $i_Q(4cs)$

$$i_Q(t) = -3.3[F] \left(\frac{dV_A(t)}{dt} \right)$$

$$i_Q(4cs) = \left(\frac{(-3.3[F])(6.05[V])}{3.826[cs]} \right) e^{1/3.826}$$

$$i_Q(4cs) = -14.9[A]$$

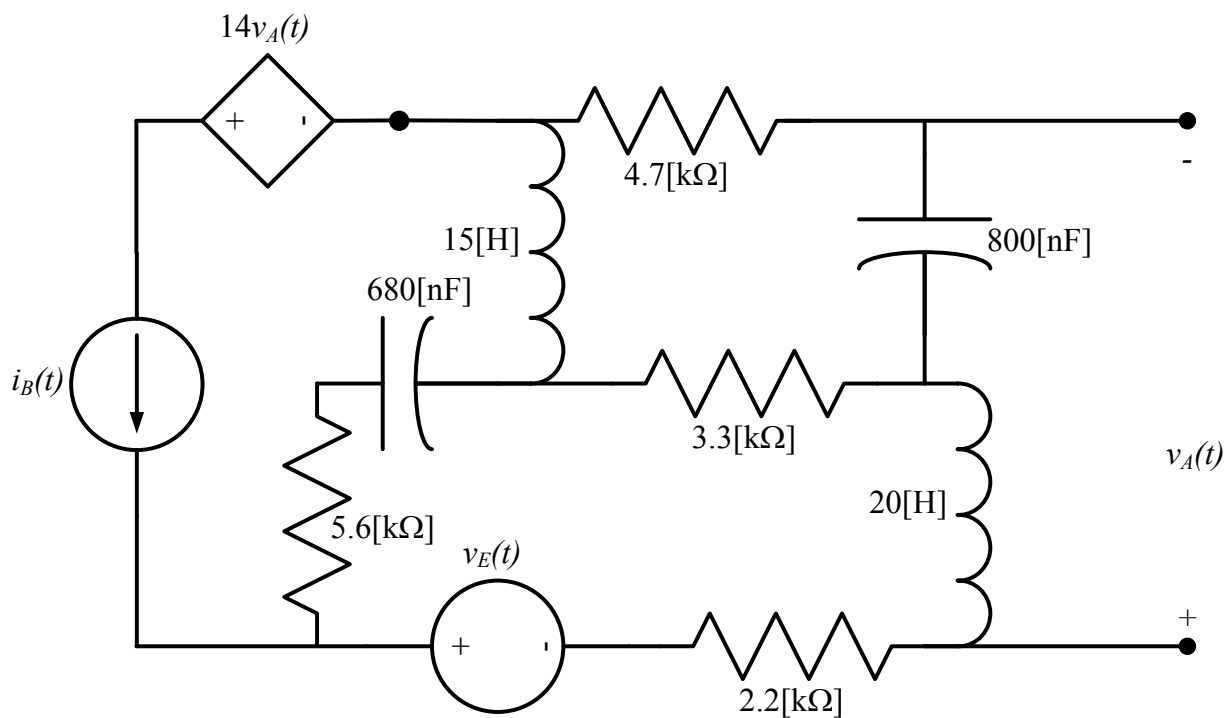
2. The circuit shown below operates in steady-state.

a) Find a numerical expression for $v_A(t)$.

b) Find the Thevenin impedance as seen by the current source, in the phasor domain.

$$v_E(t) = 47[\text{V}] \sin\left(250\left[\frac{\text{rad}}{\text{s}}\right]t + 27^\circ\right)$$

$$i_B(t) = 13[\text{mA}] \cos\left(250\left[\frac{\text{rad}}{\text{s}}\right]t - 43^\circ\right)$$



Solution in [Fall 2018, Final Exam, Problem 4](#)

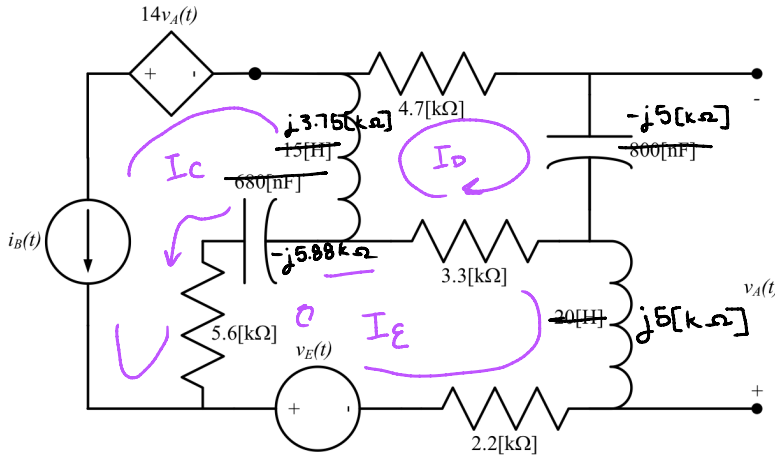
Convert:

$$v_E(t) = 47[V] \sin\left(250\left[\frac{\text{rad}}{\text{s}}\right]t + 27^\circ\right) \rightarrow 47[V] \cos(250[\text{rad/s}]t - 63^\circ)$$

$$i_B(t) = 13[\text{mA}] \cos\left(250\left[\frac{\text{rad}}{\text{s}}\right]t - 43^\circ\right)$$

$$Z_L = j\omega L \rightarrow j(250)(20) = j5[\text{k}\Omega]; j3.75[\text{k}\Omega]$$

$$Z_C = \frac{-j}{\omega C} \rightarrow -j5.88[\text{k}\Omega]; -j5[\text{k}\Omega]$$



$$V_A = (43.96 \angle -26.44^\circ)[V]$$

$$I_E = (4.892 \angle 85.47^\circ)[\text{mA}]$$

$$I_D = (4.629 \angle -139.7^\circ)[\text{mA}]$$

$$I_C = (13.0 \angle 137^\circ)[\text{mA}]$$

Part A)

$$I_B = 13 \angle -43^\circ$$

$$I_C = -13 \angle -43^\circ$$

$$I_D \rightarrow 4.7[\text{k}\Omega] I_D - j5[\text{k}\Omega] I_D + 3.3[\text{k}\Omega] (I_D - I_E) + j3.75[\text{k}\Omega] (I_D - I_C) = 0$$

$$I_E \rightarrow j5[\text{k}\Omega] I_E + 2.2[\text{k}\Omega] I_E - 47 \angle -63^\circ [V] + (I_E - I_C)(5.6 - j5.88)[\text{k}\Omega] + 3.3[\text{k}\Omega] (I_E - I_D) = 0$$

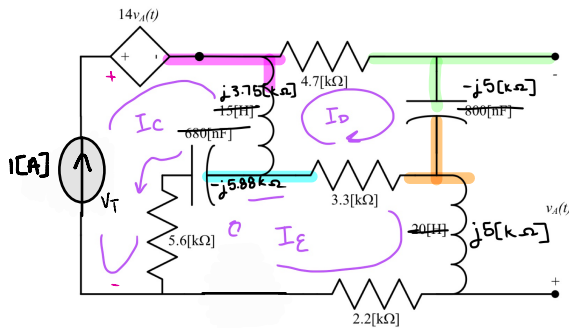
$$V_A \rightarrow -j5[\text{k}\Omega] (-I_D) - V_A + j5[\text{k}\Omega] (I_E) = 0$$

$$V_A(t) = 43.96[V] \cos(250[\text{rad/s}]t - 26.44^\circ)$$

Part B)

Since we are trying to find Thevenin, we can take the independent sources out.

Test Method:



$$I_C = 1[A]$$

$$I_D \rightarrow 4.7[\text{k}\Omega] I_D - j5[\text{k}\Omega] I_D + 3.3[\text{k}\Omega] (I_D - I_E) + j3.75[\text{k}\Omega] (I_D - I_C) = 0$$

$$I_E \rightarrow j5[\text{k}\Omega] I_E + 2.2[\text{k}\Omega] I_E - 47 \angle -63^\circ [V] + (I_E - I_C)(5.6 - j5.88)[\text{k}\Omega] + 3.3[\text{k}\Omega] (I_E - I_D) = 0$$

$$V_A \rightarrow -j5[\text{k}\Omega] (-I_D) - V_A + j5[\text{k}\Omega] (I_E) = 0$$

$$14V_A + j3.75[\text{k}\Omega] (I_C - I_D) + (5.6[\text{k}\Omega] - j5.88[\text{k}\Omega]) (I_C - I_E) - V_T = 0$$

$$V_T = (51.42 \angle -150.54^\circ)[\text{kV}]$$

$$I_C = 1[A]$$

$$I_D = (390.9 \angle 60.8^\circ)[\text{mA}]$$

$$I_E = (703.1 \angle -32.82^\circ)[\text{mA}]$$

$$V_A = (4.132 \angle -150.99^\circ)[\text{kV}]$$

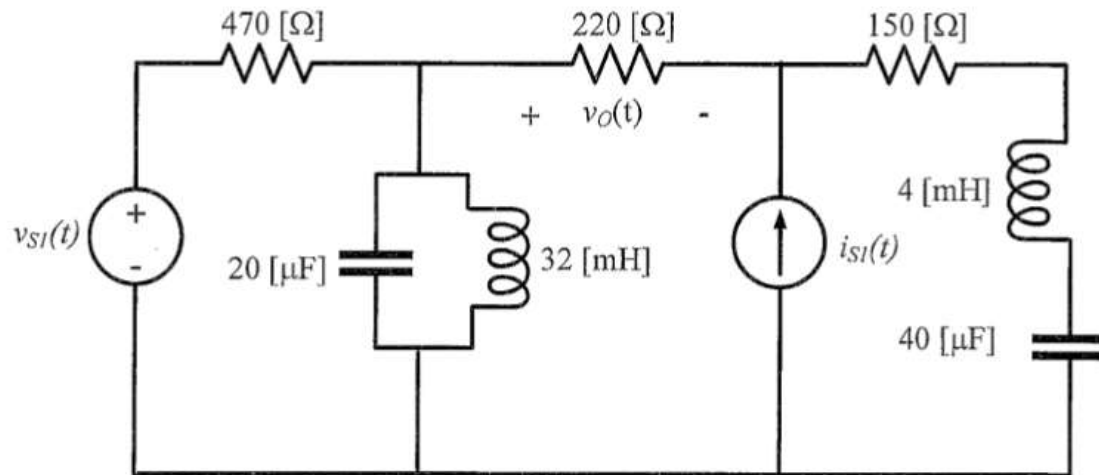
$$Z_m = \frac{V_T}{I_T} = (51.42 \angle -150.54^\circ)[\text{k}\Omega]$$

3.

5. {40 points} For the following circuit, find $v_O(t)$. It is given that

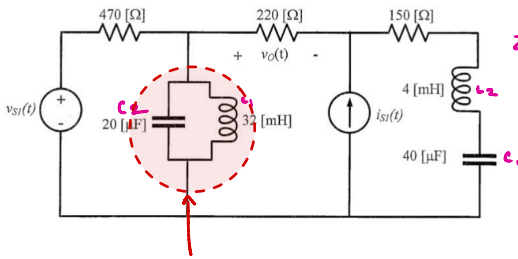
$$v_{S1}(t) = 17.5 \text{ [mV]} \cos \left(1250 \left[\frac{\text{rad}}{\text{s}} \right] t - 22^\circ \right), \text{ and}$$

$$i_{S1}(t) = 32.7 \text{ [mA]} \sin \left(2500 \left[\frac{\text{rad}}{\text{s}} \right] t + 13^\circ \right).$$



Solution in [Fall 2023, Final Exam Problem 5](#)

5. {40 points} For the following circuit, find $v_O(t)$. It is given that $v_{S1}(t) = 17.5 \text{ [mV]} \cos(1250 \frac{\text{rad}}{\text{s}} t - 22^\circ)$, and $i_{S1}(t) = 32.7 \text{ [mA]} \sin(2500 \frac{\text{rad}}{\text{s}} t + 13^\circ)$.



Note that these elements in parallel equal ∞ , making an open circuit.

Convert:

$$Z_{L1} = j\omega L \rightarrow j(1250)(32 \text{ [mH]}) = j40 \text{ [}\Omega\text{]}$$

$$Z_{L2} = j\omega L \rightarrow j(1250)(4 \text{ [mH]}) = j5 \text{ [}\Omega\text{]}$$

$$Z_{C1} = \frac{-j}{\omega C} \rightarrow \frac{-j}{1250(40 \mu\text{F})} = -j20 \text{ [}\Omega\text{]}$$

$$Z_{C2} = \frac{-j}{\omega C} \rightarrow \frac{-j}{1250(20 \mu\text{F})} = -j40 \text{ [}\Omega\text{]}$$

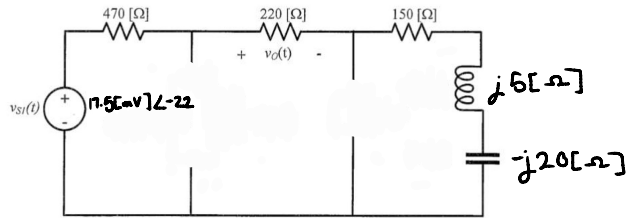
Note when dealing with 2 different freq. sources you will do the problem twice for each source.

$I_{S1}(t)$ Source

$i_{S1}(t) = V_{S1} \rightarrow$ This is Shorted $\rightarrow I_{S1} = 32.7 \angle -17^\circ \text{ [mA]}$

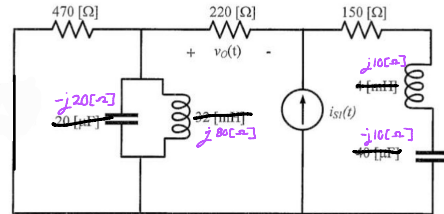
3.

5. {40 points} For the following circuit, find $v_O(t)$. It is given that $v_{S1}(t) = 17.5 \text{ [mV]} \cos(1250 \frac{\text{rad}}{\text{s}} t - 22^\circ)$, and $i_{S1}(t) = 32.7 \text{ [mA]} \sin(2500 \frac{\text{rad}}{\text{s}} t + 13^\circ)$.



$$V_O(t) = 17.5 \text{ [mV]} \angle -22^\circ \cdot \left(\frac{220 \text{ [}\Omega\text{]}}{(470 + 220 + 150 + j150 - j20) \text{ [}\Omega\text{]}} \right)$$

$$V_O(t) = 4.279 - j1.640 \text{ [mV]} \approx 4.583 \angle -20.98^\circ \text{ [mV]}$$



$4 \text{ [mH]} \rightarrow j2500(4 \text{ [mH]}) \rightarrow j10 \text{ [}\Omega\text{]}$
 $40 \mu\text{F} \rightarrow -j \rightarrow \frac{-j}{2500(40 \mu\text{F})} \rightarrow -j10 \text{ [}\Omega\text{]}$

$32 \text{ [mH]} \rightarrow j80 \text{ [}\Omega\text{]}$
 $20 \mu\text{F} \rightarrow -j20 \text{ [}\Omega\text{]}$

$$Z_{eq} = \frac{1}{470 \text{ [}\Omega\text{]}} + \frac{1}{-j20 \text{ [}\Omega\text{]}} + \frac{1}{j80 \text{ [}\Omega\text{]}} \Rightarrow 1.508 - j26.58 \text{ [}\Omega\text{]}$$

$$= (26.62 \angle -86.75^\circ) \text{ [}\Omega\text{]}$$

$$V_O''(t) = -(32.7 \text{ [mA]} \angle -17^\circ) \left(\frac{150 \text{ [}\Omega\text{]}}{(126.62 \angle -86.75^\circ) + 150 + 220 \text{ [}\Omega\text{]}} \right)$$

$$= -2.897 \angle -72.9^\circ \text{ [V]}$$

this will get $V_O(t)$ into

$$V_O(t) = -2.897 \cos(2500 \frac{\text{rad}}{\text{s}} t - 72.9^\circ) \text{ [V]}$$